Derek Chen

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MIT Masters candidate in computer science interested in problem solving and AI for science.

EDUCATION

Massachusetts Institute of Technology

Prospective M.Eng. in Computer Science — *Jan. 2026 to May 2027*

B.S. in Computer Science, B.S. in Biology — *Sep. 2022 to Dec. 2025 ◆ GPA: 5.0/5.0*

Relevant Coursework: Algorithms [6.1220], Cell Biology [7.06], Deep Learning [6.7960], NLP [6.8611], Organic Chemistry [5.13], Operating Systems [6.1810], Probability [18.600], Software Construction [6.1020], Statistics [18.650]

Experience: Graduate Teaching Assistant for Software Construction [6.102], Lab Assistant for Introduction to Algorithms [6.121] and Software Construction [6.102], Grader for Design and Analysis of Algorithms [6.122], Grader for Deep Learning [6.7960]

Skills: C++, Python, Java, JavaScript, HTML, PyTorch, TensorFlow, NumPy/Pandas

RESEARCH AND EXPERIENCE

Graduate Research Assistant at the Barzilay Group — Jan. 2024 to present

- Developing a surface point cloud-based multi-resolution autoregressive protein design model
- Developing, benchmarking, and optimizing a permutation-invariant solvent-aware protein structure diffusion model with heightened atomistic understanding (ICML GenBio 2026 Spotlight)
- Designed a variable-length ribosomal-inspired protein generative model (ICML GenBio 2025)

Graduate Research Assistant at the Zhang Group — Jan. 2025 to present

- Investigating diffusion posterior sampling for chromatin structure prediction across genomic scales

Algorithmic Development Intern at Hudson River Trading — May to Aug. 2026

- Conducted mid/high-frequency quantitative research and designed algorithmic trading strategies

Quantitative Trading Intern at Susquehanna International Group — June to Aug. 2025

- Studied options theory and designed a seasonal volatility forecasting model for commodities

Software Engineering Intern at TriNetX — May to Aug. 2024

- Developed a ML integration through proxy server forwarding utilizing the gRPC framework

Undergraduate Research Assistant at the Keating Lab — Dec. 2023 to Aug. 2024

- Developed a data generation pipeline using molecular dynamics simulations and flow matching

Machine Learning Intern at Air Force Research Laboratory — June to Aug. 2023

- Developed a machine learning pipeline to design phononic crystals with desired bandgaps

Undergraduate Research Assistant at the Liu Lab — Dec. 2022 to Sept. 2023

- Performed wet lab and computational research to predict ideal pegRNAs for prime editing

Teaching Assistant for USA Biolympiad — May to June 2023

Leadership: HMMT, Traders@MIT Officer — Sept. 2022 to present

Leadership: MIT Science Olympiad Tournament Director — Feb. 2024 to present

- Directed a Science Olympiad tournament with over 1000 attendees from across the nation

ACHIEVEMENTS AND AWARDS

International Biology Olympiad (IBO) Silver Medalist — 2021

USA Biology Olympiad (USABO) National Finals Gold Medalist — 2021 and 2022